Task 1:

In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

Answer:

Step 1: Add one Server-PT, one 2960 Switch, and two PC-PT devices.

Step 2: Connect the server and both PCs to the switch using Copper Straight-Through cables.

Step 3: Click Server → Desktop → IP Configuration. Set the static IP address to 192.168.1.1 and subnet mask to 255.255.255.0.

Step 4: Open Server → Services → DHCP. Turn DHCP On and configure the following settings:

- Pool Name: LAN_POOL
- Start IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Maximum Users: 50

Click Add.

Step 5: Open PC0 → Desktop → IP Configuration and select DHCP.

Step 6: Repeat the same process for PC1 by selecting DHCP.

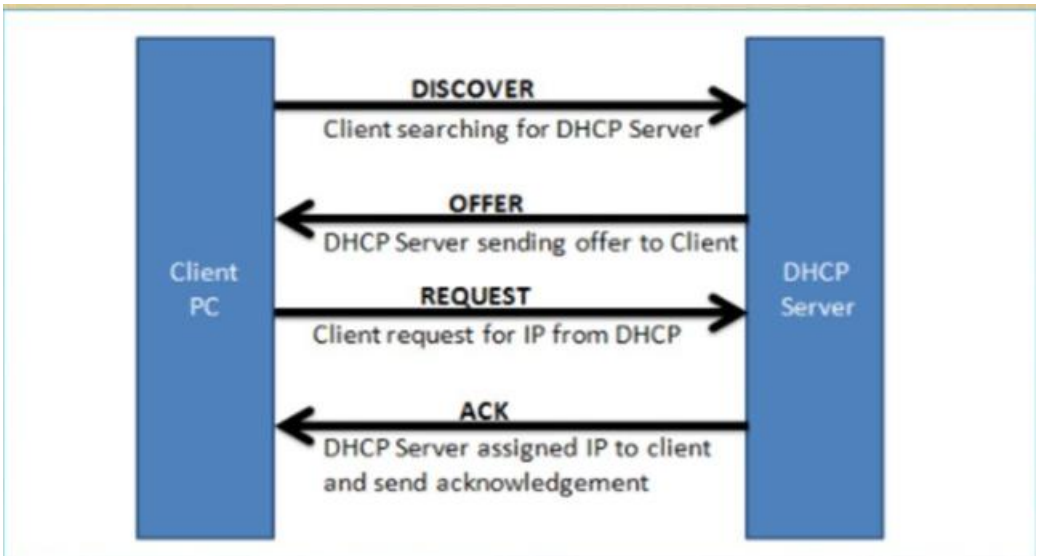
Step 7: Open Command Prompt on both PCs and run ipconfig to verify that they have different IP addresses.

Step 8: From PC0, run ping followed by PC1's IP address to test connectivity.

Task 2:

Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.

Answer:



DHCP Process – Four Steps (DORA)

1. Discover: The client sends a DHCP Discover message to find a DHCP server.

2. Offer: The DHCP server sends a DHCP Offer message with an available IP address to the client.

3. Request: The client sends a DHCP Request message to request the offered IP address.

4. Acknowledge (ACK): The DHCP server sends a DHCP ACK message to confirm the IP address assignment.

Task 3:

Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

Answer:

1. Open Cisco Packet Tracer and create a network with one DHCP server, one switch, and two PCs.
2. Connect the server and both PCs to the switch using Copper Straight-Through cables.
3. Click the server and open Desktop → IP Configuration. Set the IP address to 192.168.1.1 and the subnet mask to 255.255.255.0.
4. Open Services → DHCP on the server and turn DHCP On.
5. Configure the DHCP pool with the following settings:
    - Pool Name: LAN_POOL
    - Start IP Address: 192.168.1.10
    - Subnet Mask: 255.255.255.0
    - Maximum Number of Users: 1
6. Click Add or Save to apply the DHCP pool settings. Ensure that only one IP address is available in the pool.
7. Click PC0, open Desktop → IP Configuration, and select DHCP. The PC should receive the IP address 192.168.1.10.
8. Click PC1, open Desktop → IP Configuration, and select DHCP. Since the only available IP address has already been assigned to PC0, PC1 will be unable to obtain another valid IP address.
9. Check the IP configuration of both PCs to observe the result.

Result and Explanation:

The DHCP server successfully assigns the available IP address 192.168.1.10 to one PC. The second PC cannot obtain an IP address because the DHCP pool has only one available address, which is already assigned to the first PC. Therefore, only one PC can receive an IP address automatically until another address is added to the DHCP pool.

Task 4:

Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

Answer:

Steps to Perform in Cisco Packet Tracer

1. Add devices: Add one Server-PT, one Wireless Router, one Laptop-PT, and one generic PC-PT to represent a smartphone.

2. Configure the DHCP server: Click Server → Desktop → IP Configuration. Set the IP address to 192.168.1.2 and subnet mask to 255.255.255.0.

3. Enable DHCP: Open Server → Services → DHCP, turn DHCP On, and configure the following settings:

- Pool Name: Zomato_WiFi
- Start IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- Maximum Number of Users: 50

Click Add to save the pool.

4. Configure the wireless router: Set its LAN IP address to 192.168.1.1 and configure a Wi-Fi SSID such as Zomato_Restaurant with a password. Disable the router's own DHCP service to avoid conflicts.

5. Connect the server: Connect the server to a LAN port on the wireless router using a Copper Straight-Through cable.

6. Connect the laptop: Open Laptop → Desktop → PC Wireless, select Zomato_Restaurant, and connect using the Wi-Fi password. Then open Desktop → IP Configuration and select DHCP.

7. Connect the smartphone: Use the generic PC as the smartphone. If the PC has a wireless interface, connect it to the same Wi-Fi network. Otherwise, use a wireless-capable device or install a compatible wireless module if supported by the device. Select DHCP in its IP Configuration settings.

8. Verify the IP addresses: Check the IP configuration of both clients. They should receive different addresses from the DHCP pool.

Expected Result

Device	Example assigned IP address
Laptop	192.168.1.10
Smartphone (generic PC)	192.168.1.11

Why is DHCP useful in a Zomato-style restaurant?

DHCP automatically assigns IP addresses to customer devices when they connect to the restaurant's Wi-Fi. It reduces manual configuration, prevents duplicate IP address assignments, and makes it easier to manage many customer devices.

Important: This setup assumes the server is providing DHCP on the same LAN as the wireless router. Ensure that the router's DHCP service is disabled and that the clients can reach the server; otherwise, the clients may not receive addresses from the intended pool.